

PERSONAL LEARNING JOURNEY

Member 2 – Jayamuni J T S J – IT24300329
Assigned role: EDA and Data-Quality Analysis

My contribution

My job was to explore the dataset and check its quality before any model was built. I wanted to find out which patterns are real and which are only chance. I wrote four R scripts: one shows how the dataset was built, one describes the data, and two run statistical tests. I also made a notebook that looks at other patterns, such as player nationality and red cards. I recorded 20 findings in an insight log. Each finding says what we asked, what we found, and what it means for the model.

What I learned

I learned that exploring data is only useful when it changes a decision. Home teams win 45.0% of matches, 24.9% are draws and 30.1% are away wins. Because draws are the smallest group, the group chose a score (Macro-F1) that treats all three results equally. I also found that results change slowly over the years, so we trained on older seasons and tested on newer ones, never mixing them at random. Another lesson: a missing value does not always mean zero. In the first match of a season a team has no recent form yet, and that is not the same as bad form.

Challenge

Our dataset is large, with about 42,000 matches in the training period. With so much data, almost every test says a result is "significant", so that word alone could not tell us what matters. Many features were also unevenly spread, which breaks the rules of some standard tests. Some patterns looked strong but could not be used. Teams that get a red card lose 57.1% of their matches, compared with 36.7% without one. But a red card happens during the game, so it cannot be known before kickoff.

Solution

I ran every test on the training seasons only (up to 2022), so the test seasons stayed untouched. I used tests that work with uneven data, and I ranked features by the size of their effect, not only by whether they were "significant". The difference in starting-XI strength between the two teams had the largest effect (0.145), about twice that of recent form (0.06 to 0.08). Midfield strength and attack against defence came next, and keeping the same line-up showed no link at all ($p = 0.57$). Later, the models also found starting-XI strength to be the most useful feature.

Reflection

Exploring data is more than drawing charts. Writing "what this means for the model" next to every finding made me connect my results to the group's decisions. It also made it easy to hand my results to the members working on features and models. I now understand that "statistically significant" and "important" are two different questions, especially with a large dataset.

Future improvement

I would study draws more closely, because they are the hardest result to predict. I would also compare home advantage league by league around the COVID season. With empty stadiums, home wins fell to 40.2%, compared with 45.7% in the other seasons. Finally, I would share my findings with the feature team earlier, so that new ideas could be tested sooner.